Characterization of two-dimensional electron gas in amorphous indium gallium zinc oxide (IGZO) thin film transistor

Alessandro Ciavatta, Elisa Damiani, Riccardo Gaeta

June 2022

Abstract

An IGZO TFT output and transfer characteristics are investigated. From the transfer characteristics the FET mobility is obtained in the two regimes, $\mu_{lin} = (6.89 \pm 0.19)\,\mathrm{cm}^2/\mathrm{Vs}$ for the linear and $\mu_{sat} = (6.3 \pm 0.1)\,\mathrm{cm}^2/\mathrm{Vs}$ for the saturation regime. The threshold voltage is determined to be $V_{th} = 0.4 \pm 0.2\,\mathrm{V}$ via the output characteristics. The ON current observed is $I_{ON} = 23\,\mu\mathrm{A}$ at $V_G = 5\,\mathrm{V}$, while the average current in the OFF regime is $I_{OFF} = 27 \pm 13\,\mathrm{pA}$ granting a ratio I_{ON}/I_{OFF} in the order of 10^6. The voltage at which an onset in channel conductivity occurs is $V_{ON} = -1.5 \pm 0.1\,\mathrm{V}$ and the subthreshold slope value is found to be $S = 0.283 \pm 0.009\,\mathrm{V/dec}$.

Introduction

Thin film transistors (TFT), as the name suggests, are a type field-effect transistors (FET) with a length and width much larger than their height that are built atop a substrate in a thin film-like layer. TFTs containing a layer of amorphous oxide semiconductor (AOS) are widely used in industry to easily cover large areas like displays or detectors. The most commonly used AOS is indium gallium zinc oxide (IGZO) that behaves like a n-type semiconductor and has the advantage over amorphous Si thanks to its higher electron mobility [1].

An AOS TFT is fabricated on an inert substrate by physical deposition of various nm-thick layers as shown in the the schematics of a generic TFT in Fig.(1): first the metallic gate (G) contacts are deposited, then a dielectric (usually SiO_2) followed by the AOS semiconductor and finally the source (S) and drain (D) contacts.

With the right choice of substrate and dielectric, transparent and/or flexible TFT-based devices can be built.

By applying a positive voltage to the gate V_G (in reference to S) a thin conductive channel is formed at the interface between the IGZO semiconductor and the dielectric allowing for the possibility of a current to flow between the source and drain.

The electric field responsible for the creation of the channel tightly confines the electrons at the interface in a very thin layer that can be described as a 2-D electron gas (2DEG)

By characterizing such a transistor the transport properties of this 2DEG can be probed. This is the objective of this work.

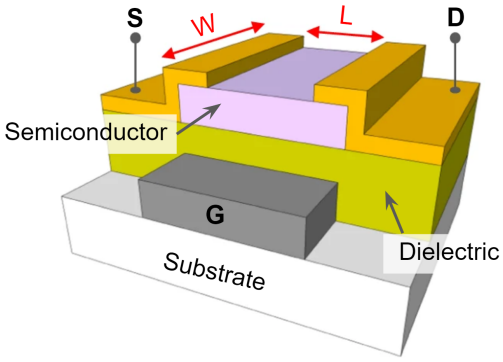


Figure 1: Schematic representation of a thin film transistor

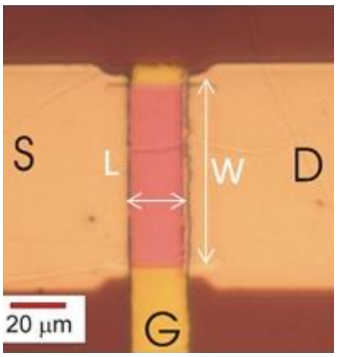


Figure 2: Microscopy top view of a similar device that will be studied in this work

The conductive channel is only formed when charges start accumulating at the interface, which happens when the gate voltage surpasses a threshold voltage $V_G > V_{th}$. All the voltages in this work are to be interpreted as in reference to the source S potential.

It is possible to distinguish between two main working regions of the transistor: linear and saturation regimes.

Linear regime

The FET is in the linear regime when the applied voltage V_D satisfies the condition: $V_D < V_G - V_{th}$. In this regime, the potential at a generic point of the channel is given by a function $V(x)$ that varies linearly from the source ($x = 0$) to the drain ($x = L$) where the potential is V_D:

$$V(x) = \frac{V_D}{L}x.$$

The current flowing in D, ignoring the thickness, is

$$I_D = W\sigma_{avg}v_{avg}$$

where W refers to the width of the channel (Fig.(2)) the subscript avg refers to the average over the length L of the channel. The surface charge density σ is defined as:

$$\sigma(x) = \frac{Q}{A} = c[(V_G - V_{th}) - V(x)]$$

where $c = \frac{C}{A}$ is the specific capacitance of the dielectric. The speed of the charge carriers v is:

$$v(x) = \mu E(x) = \mu\frac{\partial V(x)}{\partial x} = \mu\frac{V_D}{L}$$

where μ is the electron FET mobility. The average of these two quantities are:

$$\sigma_{avg} = \frac{1}{L}\int_0^L \sigma(x)dx = c\left[(V_G - V_{th}) - \frac{V_D}{2}\right]$$

and

$$v_{avg} = \frac{1}{L}\int_0^L v(x)dx = \mu\frac{V_D}{L}$$

The expression for the current flowing between S and D in the linear regime is thus:

$$I_D = \mu c\frac{W}{L}\left[(V_G - V_{th})V_D - \frac{1}{2}V_D^2\right] \qquad (1)$$

By introducing the transconductance g_m,

$$g_m = \left(\frac{\partial I_D}{\partial V_G}\right)_{V_D} = \mu\frac{cWV_D}{L}$$

it is easy to determine the FET mobility in the linear regime as:

$$\mu_{lin} = \frac{L}{cWV_D}\left(\frac{\partial I_D}{\partial V_G}\right) \qquad (2)$$

Saturation regime

At $V_D = V_G - V_{th}$, the charge density close to D falls to zero (pinch off condition), thus changing the behaviour of the transistor: I_D saturates.

The condition $V_D > V_G - V_{th}$ defines the saturation regime of the transistor: an increase of V_D slightly extends the region void of charges (the depletion region) around the drain, without leading to a significant increase of I_D. In this case the expression for I_D becomes:

$$I_D = \mu c\frac{W}{L}\frac{1}{2}(V_G - V_{th})^2 \qquad (3)$$

The FET mobility in the saturation regime can thus be obtained:

$$\mu_{sat} = \frac{2L}{cW}\left(\frac{\partial\sqrt{I_D}}{\partial V_G}\right)^2 \qquad (4)$$

These are two method of valuating the mobility inside the FET. The mobility calculated from the linear regime μ_{lin} tends to overestimate the real μ, while μ_{sat} tends to underestimate it [2].

Materials and methods

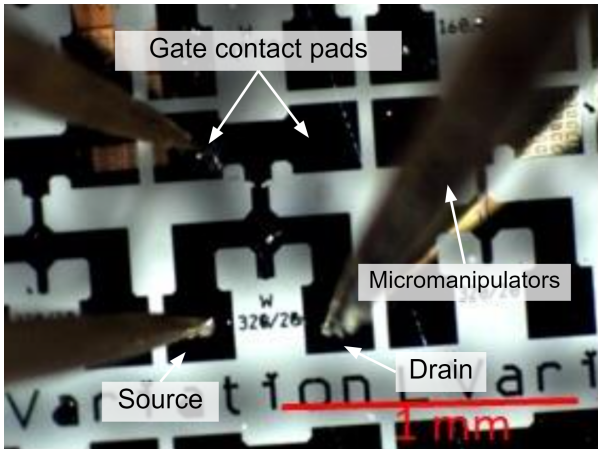


Figure 3: Microscope image of an equivalent TFT mounted inside the probestation

The AOS TFT that has been characterized is made of a dielectric oxide layer that contains Ta_2O_5 mixed with SiO_2, a semiconducting oxide layer of IGZO and three conductive molybdenum electrodes.

The channel dimesions are $L = 20\,\mu m$ and $W = 320\,\mu m$, while the specific capacitance of the dielectric layer is assumed to be exactly $50\,nF/cm^2$. The DUT, shown in Fig.(3), is put inside a probestation and, with the help three micromanipulators, its electrodes are connected to a Agilent B2900 [3] source/measure unit (SMU). To measure accurately the small currents flowing in this device (as low as 10^{-11}A) the micromanipulators attached to the G and D electrodes are connected by triaxial cables to the HI-ports of the two channels of the SMU, while the S electrode is connected to the LO-ports using a coaxial cable.

The SMU is software controlled and lets the user acquire data using automatic potential sweeps, a feature necessary to acquire transfer and output characteristics of the device.

The output characteristics I_D vs. V_D for different V_G has been acquired by performing a sweep of the voltage V_D from 0 to 9 V, for different constant V_G values that go from -1 to 5 V with a step of 1 V.

The transfer characteristics of I_D vs. V_G has been acquired by sweeping V_G from -5 V to 5 V and back again, for different constant values of V_D from 1 to 8 V, with a 1 V step. From this graph both the threshold voltage and the mobilities in the two regimes can be calculated. To obtain the voltage threshold two methods can be used: V_{th} is given by the intersection between the linear fit and the $I_D = 0$ axis of either $I_D(V_D)$ for a low value of V_G (linear regime) or of $\sqrt{I_D(V_D)}$ for a high value of V_G (saturation regime) [4]. These two values probably overestimate and underestimate the true V_{th} and so they are averaged.

Results and discussion

The output characteristics of the IGZO TFT is shown in Fig.(4). The red parabolic line represent the $V_D = V_G - V_{th}$ condition that delimits the linear from the saturation regime of the transistor. For negative values of V_G the current becomes negligible. All the acquired characteristics show an hysteresis effect of uninvestigated origins that seems to increase with V_G.

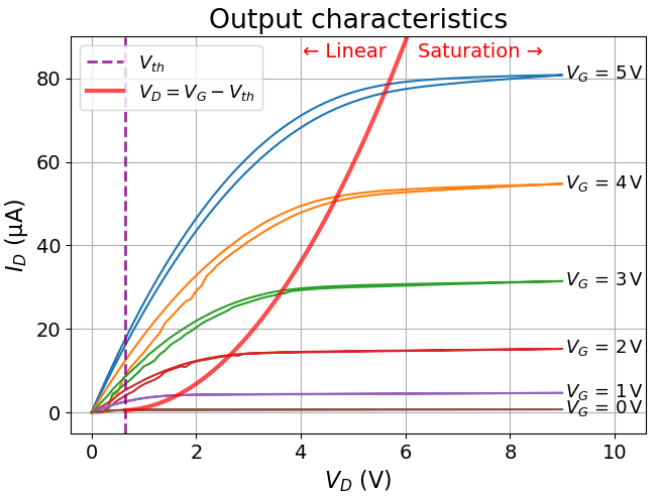


Figure 4: Output characteristics. The red parabolic line represents the condition that divides linear from saturation regime $V_D = V_G - V_{th}$. Each I-V plot is acquired for different values of V_G

The transfer characteristics and their respective linear fit are acquired for $V_D = 1$ V and $V_D = 8$ V and are shown in Fig.(5). The two values of calculated V_{th} are summarized in Tab.(1). The two values are discordant, but nonetheless the average threshold value is calculated.

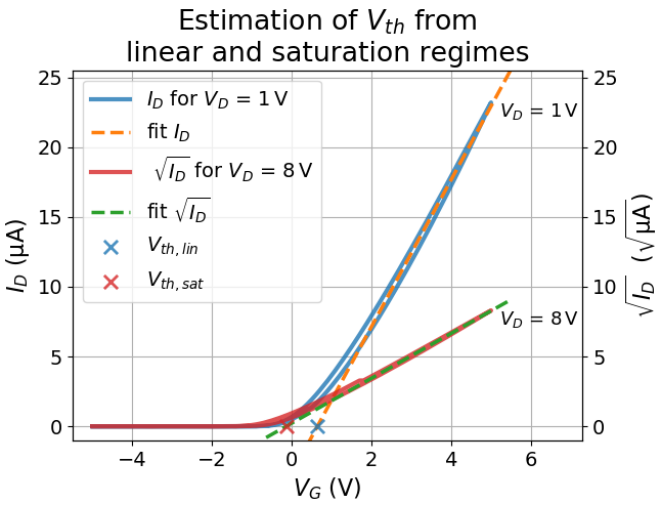


Figure 5: Transfer characteristics with their linear fit. The blue line is the behaviour of I_D for $V_D = 1\,\text{V}$, while the red one is the one of $\sqrt{I_D}$ for $V_D = 8\,\text{V}$. The two crosses represent V_{th} calculated for the two regimes

	V_{th} (V)	R^2
Linear	0.64 ± 0.04	0.992
Saturation	-0.1 ± 0.2	0.996
Average	0.4 ± 0.2	—

Table 1: Threshold voltage calculated from linear and saturation regimes

By using the slope of the fit in Eq.(2) and Eq.(4) for the linear and saturation case the FET mobilities are obtained : for the linear regime μ_{lin} is $(6.89 \pm 0.19)\,\text{cm}^2/\text{Vs}$ while for the saturation one $\mu_{sat} = (6.3 \pm 0.1)\,\text{cm}^2/\text{Vs}$.

In theory the two values should be the same, but in real devices the assumption that the mobility remains constant over all ranges of applied V_G is not correct, as it depends on the distribution of the charge within the channel. These results are close to the value found in literature [5].

The transfer characteristics of $\log(I_D)$, only for the $V_D = 1\,\text{V}$ case, is shown in Fig.(6). By performing a linear fit it is possible to derive the subthreshold slope S, which indicates the sharpness of the switching behavior of the 2D FET [6]. The obtained result is $S = 0.283 \pm 0.009\,\text{V/dec}$, which is again compatible with the values found in literature [1]. The estimation of the transistor parameter I_{ON} is given by the maximum value of I_D, which in our case is at $V_G = 5\,\text{V}$: $I_{ON} = 23.2\,\mu\text{A}$.

The value of I_{OFF} has been calculated by performing an average of I_D from -5 V up to -2 V. The result is $I_{OFF} = 27 \pm 13\,\text{pA}$.

To conclude, V_{ON}, the voltage at which the channel conductivity behaviour drastically changes, has been determined from the intersection of the linear fit and the line corresponding to I_{OFF}, which gives $V_{ON} = -1.5 \pm 0.1\,\text{V}$.

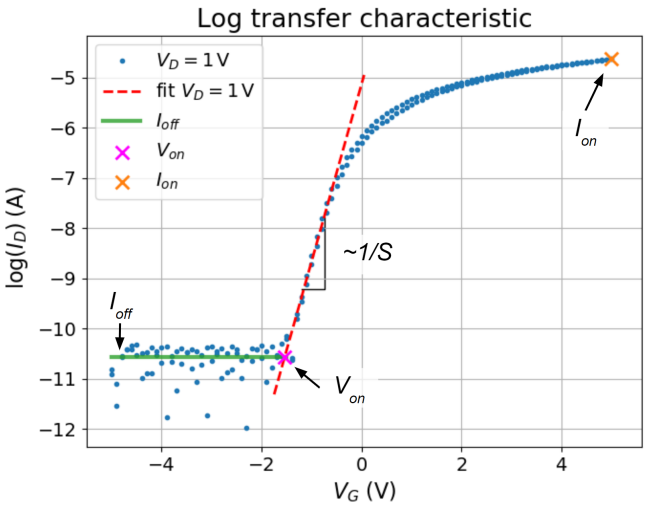


Figure 6: Transfer characteristics of $\log I_D$ for $V_D = 1\,\text{V}$

Conclusions

By using a probestation and sensitive measuring equipment it is possible to characterize a single IGZO TFT and to extract the transport properties of the 2DEG contained in the transistor channel.

From the transfer characteristics in the two regimes the calculated FET mobilities are: $\mu_{lin} = (6.89 \pm 0.19)\,\text{cm}^2/\text{Vs}$ and $\mu_{sat} = (6.3 \pm 0.1)\,\text{cm}^2/\text{Vs}$. The threshold voltage is $V_{th} = 0.4 \pm 0.2\,\text{V}$. The values for the transistor parameters I_{ON} and I_{OFF} are respectively $I_{ON} = 23.2 \pm 0.5\,\mu\text{A}$ and $I_{OFF} = 27 \pm 13\,\text{pA}$, leading to a I_{ON}/I_{OFF} ratio of the order of 10^6. The voltage at which an onset in channel conductivity occurs is $V_{ON} = -1.5 \pm 0.1\,\text{V}$. Lastly, the obtained result for the subthreshold value is $S = 0.283 \pm 0.009\,\text{V/dec}$.

References

[1] Toshio Kamiya1 and Hideo Hosono. "Material characteristics and applications of transparent amorphous oxide semiconductors". In: *NPG Asia Matereials* 2.1 (2010), pp. 15–22. DOI: 10.1038/asiamat.2010.5.

[2] Constance Rost-Bietsch. *Ambipolar and Light-Emitting Organic Field-Effect Transistors*. URL: https://books.google.it/books?id=XxvtOCkVKaIC&pg=PA19&redir_esc=y#v=onepage&q&f=false.

[3] Agilent B2900 Series SMU. URL: https://assets.testequity.com/te1/Documents/pdf/B2900A-ug.pdf.

[4] L. Dobrescu et al. "Threshold voltage extraction methods for MOS transistors". In: *2000 International Semiconductor Conference. 23rd Edition. CAS 2000 Proceedings (Cat. No.00TH8486)*. Vol. 1. 2000, 371–374 vol.1. DOI: 10.1109/SMICND.2000.890257.

[5] Shinpei Matsuda, Yasumasa Yaman, and Yutaka Okazaki et al. "Reduction in Mobility Difference between C-Axis-Aligned Crystalline IGZO-FET and Si-FET by Miniaturization". In: *2014 International Conference on Solid State Devices and Materials* (2014), pp. 138–139. DOI: 10.7567/SSDM.2014.PS-6-10.

[6] S. B. Mitta and M. S. Choi et al. "Electrical characterization of 2D materials-based field-effect transistors". In: *2D Materials* 8 (2021). DOI: 10.1088/2053-1583/abc187.